

MEDRADAR · PUBLIC HEALTH LOGISTICS NETWORK

NH-46 Healthcare Corridor Surveillance Specification (Bhopal – Sehore – Ashta – Dewas – Indore)

1. EXECUTIVE SUMMARY & PURPOSE

MedRadar is an institutional-grade, closed-loop pharmaceutical logistics and real-time stock surveillance system engineered for the Madhya Pradesh NH-46 healthcare corridor. The platform provides continuous visibility into essential drug availability across 245 verified retail pharmacies, Jan Aushadhi Kendras (PMBJP), and government hospital buffer stores.

By unifying citizen demand telemetry, pharmacist stock reporting, and distributor logistics advisories, MedRadar mitigates localized stockouts, eliminates geographical disparities in essential drug access, and provides district health coordinators with predictive depletion velocity metrics.

2. PRODUCTION TECHNOLOGY STACK

- **Frontend Layer:** Next.js 15.1 (App Router), React 19, TypeScript, Vanilla Tailwind CSS (Slate Theme)
- **Spatial Mapping:** Leaflet 1.9.4 with CartoDB Voyager Raster Tiles & Hardware GPS Geolocation API
- **Database & Sync:** Supabase PostgreSQL (245 Facilities, 11,330 SKUs, Real-Time WebSocket Channel Engine)
- **AI & NLP Engine:** Gemini 2.5 Flash Autonomous Entity Extraction & Prompt-Injection Defense Architecture
- **Messaging & SMS:** Twilio WhatsApp Webhook API & Fast2SMS High-Throughput Regional Gateway Pipeline
- **Production Host:** Vercel Enterprise Edge Network with Global CDN & Zero-Cold-Start Serverless Lambdas

3. CORE FUNCTIONAL MODULES

Citizen Medicine Finder (Web /)

Spatial Haversine inventory matching, live driving distance calculation, 30-minute prescription hold reservation, and automated SMS restock queue registration.

Autonomous WhatsApp Concierge

Multilingual entity parser detecting drug queries, GPS pins, Google Maps URLs, prescription photos, and pharmacist stock broadcast commands ("Insulin YES" / "KHATAM").

Directorate Surveillance Dashboard (/dashboard)

Corridor deficit heatmaps, live telemetry streams, shortage depletion velocity matrix, and 1-click SMS logistics advisory dispatch to regional C&F depots.

Clinical & Pricing Dossier (/api/medicine-info)

Indian Pharmacopoeia (IP) composition standards, dosage administration guidelines, cold-chain specifications, and PMBJP Jan Aushadhi generic price comparisons (50%-84% savings).

4. CLOSED-LOOP STAKEHOLDER ARCHITECTURE

1. Patient Query: Citizen searches via GPS or WhatsApp. If 0 retail stock exists, the shortage is logged into the surveillance database.
2. Coordinator Alert: The Surveillance Dashboard aggregates deficits, alerting health administrators.
3. Distributor Dispatch: Logistics advisories trigger buffer stock reallocation from industrial depots to rural CHCs.
4. Chemist Inflow: Pharmacist marks medicine IN STOCK on WhatsApp.
5. Patient Restock Broadcast: System auto-dispatches SMS alerts to all waiting patients within a 10 km geofence.